

The Two Ratios in GRPO

A frequent source of confusion is that GRPO can contain two different probability ratios for two different jobs.

Quantity	Comparison	Purpose
Policy ratio	$\pi_{\theta} / \pi_{\text{old}}$	PPO-style clipping; controls how much the policy changes relative to the policy that generated the rollout
KL ratio / estimator	$\pi_{\text{ref}} / \pi_{\theta}$ (in the common estimator form)	Penalizes divergence of the updated policy from the frozen reference model

Unbiased per-token KL estimator used in the DeepSeekMath formulation

Let $r_t = \pi_{\text{ref}}(\text{token}_t | \text{context}) / \pi_{\theta}(\text{token}_t | \text{context})$. The per-token estimator can be written as:

$$\text{KL_estimate}_t = r_t - \log(r_t) - 1$$

Under sampling from π_{θ} , the expectation of this estimator gives the KL direction $D_{\text{KL}}(\pi_{\theta} || \pi_{\text{ref}})$. The exact notation can vary with how the ratio is defined, so always check which distribution is in the numerator.

Do not memorize the ratio without its definition. If you instead define $r_t = \pi_{\theta} / \pi_{\text{ref}}$, the algebraic form of the estimator must be written consistently.

Reward shaping vs objective regularization

Reward-shaped description	DeepSeekMath-style objective description
Modify scalar reward first, e.g. $r' = r - \beta \cdot \text{KL}$, then normalize r' across the group.	Compute the task/verifier rewards, construct group-relative advantages, then include KL regularization directly in the policy objective.
This changes the quantity being normalized for the group baseline.	The KL penalty remains an explicit policy-optimization term rather than being silently folded into the group reward.

Important scope: this wording is specifically about the DeepSeekMath-style GRPO formulation. Different GRPO implementations can make engineering choices differently, so the implementation or paper should be checked before generalizing.